When Direct Manipulation Becomes a Guess: Productive Friction in AI-Mediated Multisensory Visualization

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Abstract

Generative visualization increasingly embeds large language models within direct manipulation and multisensory interaction. Speech, gaze, touch, gesture, sound, and haptics can make probabilistic inference feel like familiar, deterministic tool use. I call this gap a *deterministic-affordance mismatch*: deterministic interaction cues carry over from users' experience with prior systems, while AI may weaken predictability, locality, reversibility, or provenance. Reading the malleable interfaces of *Iron Man 2* against systems from *Blade Runner 2049*, I propose *productive friction*, including cross-sensory renderings whose detail reflects model uncertainty. Four frictions expose the inference boundary, match action to effect, make stochastic branches tangible, and attribute sensory agreement.

Index Terms: Generative Visualization, Multisensory Interaction, Direct Manipulation, Human–AI interaction.

1 Introduction

In an *Iron Man 2* scene, Tony Stark speaks, points, and expands a projected city model while JARVIS supplies the missing computation [7]. The artefacts persist, yet an intelligent agent interprets each incomplete action. Current generative visualization resembles this fiction. DynaVis turns language into persistent widgets [17]; Data Formulator delegates transformation to an AI agent [18]; and InterChat combines direct manipulation with LLM-based intent inference [3]. They motivate a question: when does a probabilistic model adopt a deterministic tool's perceptual grammar?

Science-fiction interfaces act as diegetic prototypes shaping technological expectations [9]. They expose assumptions technical accounts leave implicit [4]. I identify this mismatch and four productive frictions that make consequential mediation perceptible.

2 A Direct Action Is a Promise

Direct manipulation rests on continuous representation, physical action, rapid feedback, and incremental, reversible operations [16]. The idea of mimetic action places responsibility on the user to control physical actions with precision, while emphasizing predictability and rapid, incremental, reversible operations on the system-side whose impact is immediately visible. The feeling of directness depends on short semantic and articulatory distances between intention, action, and result, on top of the relationship between input and output vocabularies for an interface [8]. Multimodal systems complicate this contract. They combine complementary signals, using one mode to resolve another's ambiguity [13]. A gaze can identify an object while speech names an operation. This can broaden the gulf of evaluation [11], as the impacts of user actions may not be immediately perceived.

Figure 2 separates two directions of communication. Multimodal input combines user-to-system signals—speech, gaze, touch, and gesture—to express intent. Multisensory output combines system-to-user representations—graphics, language, sound, and haptics—to convey results. Between them, the model fuses input and infers intent, generates an analytical transformation, and renders its result. Each friction sits where a direct-manipulation promise first becomes vulnerable: inference threatens predictability; transformation scope threatens locality; generated state changes threaten reversibility; and coordinated outputs threaten provenance. Thus, one small gesture can cause global change, undo can regenerate rather than restore, and several agreeing senses can repeat one model error.

Multisensory output can strengthen that feeling. People integrate visual and haptic estimates by apparent reliability [5]; haptics can encode data through force, texture, or vibration [14]. A single model can coordinate graphics, narration, and haptic emphasis, making one error feel like several reinforcing sensations. Exact-looking marks, tones, or haptic events can also overstate a coarse inference. Designers can use purposefully low-fidelity sensory representations to communicate uncertainty in model output. A coarse output could appear as a visual interval, a sonified band, or a diffuse haptic region. Each replaces an exact mark, pitch, or detent. The representation's perceptual resolution should match the degree of uncertainty introduced by the stochastic model. Exact source measurements should remain available for inspection. No sense is inherently low fidelity; designers choose the detail encoded within it.

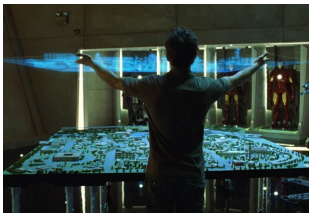
3 Two Lessons from *Blade Runner 2049*

Blade Runner 2049 offers two lessons. First, keep the evidentiary path inspectable. K, a police investigator, searches birth records for evidence of a replicant birth. The Denabase lets him advance through records and compare a suspicious duplicated pair. In a separate morgue scene, dials and optical transitions reveal a progression from bone to microscopic serial number. Both interfaces preserve the intermediate states through which an observation becomes a finding. Their material controls provide no inherent truth, but they make comparison and analytical labour visible. The production team describes these systems through physical, optical, microfiche, and mechanical references [6].

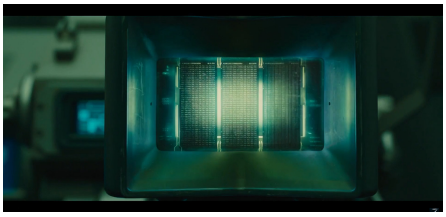
Second, preserve provenance when sensory channels align. Joi is a projected AI companion; Mariette is an embodied replicant. During their encounter, Joi's image and voice align with Mariette's body, letting K perceive one intimate presence. Momentary visual misalignments reveal that sight, touch, speech, and agency come from different sources. The lesson for AI interfaces is to disclose common origin when graphics, narration, sonification, and haptics repeat one model output. This disclosure concerns evidentiary provenance, not personhood; it must never become a sensory test for recognition. The interface should "reappear" when transparency creates epistemic risk [19].

4 Four Frictions for Four Weakened Promises

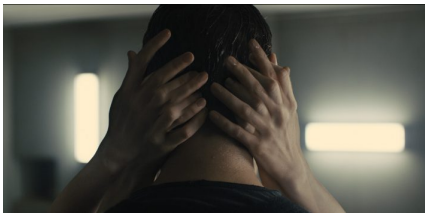
Each friction answers one failure point in the deterministic-affordance mismatch. MacLean describes haptics as making digital



(a) Malleable model



(b) Inspectable records



(c) Mixed embodiment

Figure 1: Three science-fiction scenes expose different interface promises. (a) JARVIS interprets Stark's expansive gesture over a malleable model. (b) A backlit Denabase card keeps record comparison physically visible. (c) Duplicated hands in the Joi–Mariette merge disclose multiple embodied sources. Panels (a–b) are visualization cases; panel (c) is an ethical analogy.

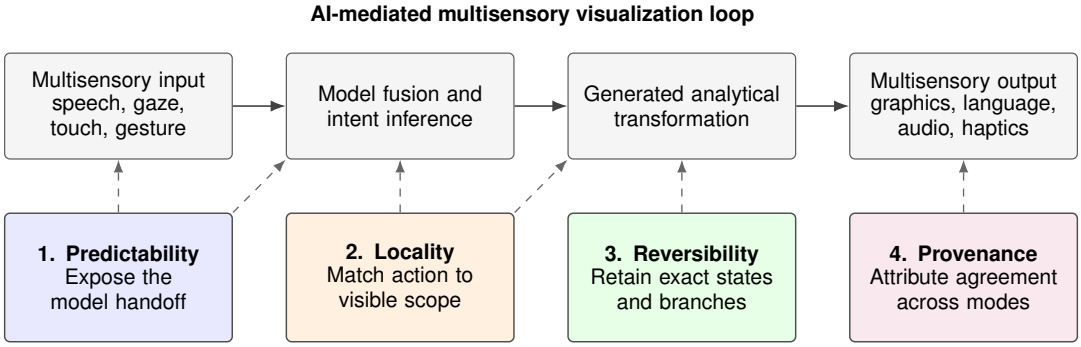


Figure 2: An AI model can mediate the full multisensory visualization loop. Each friction answers a direct-manipulation promise weakened by model mediation: predictability, locality, reversibility, or provenance. At the output stage, sensory detail can also reflect model uncertainty.

transactions physical and confirmed [10]; I extend this principle to mark the boundary between direct input and model inference.

1. Predictability—expose inference. JARVIS completes Stark's partial speech and gesture. A real interface should expose that handoff before execution, consistent with human–AI guidance [1]. It can ghost the inferred target, operation, and cross-modal binding; a haptic detent, auditory transition, or visual threshold can separate exploration from commitment.

2. Locality—match action to effect. A small gesture should not silently trigger a global transformation. Before execution, a scope halo should reveal every affected dataset, view, and encoding. Effects exceeding the action's apparent scale require an explicit second commitment. An inspectable trace exposes generated code, assumptions, and downstream changes.

3. Reversibility—make branches perceivable. The Denabase and morgue sequences keep intermediate evidence visible as K advances. Generative interfaces should likewise preserve exact checkpoints and expose alternative histories. Undo restores the prior state; replay reuses captured inputs and settings; regenerate creates a labelled stochastic branch. Persistent visual, auditory, or tangible tokens can support comparison between histories.

4. Provenance—make agreement between senses explicit. The Joi–Mariette analogy keeps several embodied sources visible despite aligned outputs. AI interfaces should likewise attribute agreement across graphics, narration, sonification, and haptic emphasis. Coordinated cues need a common-source marker to avoid implying independent sources of knowledge. At the same time, measured data, external evidence, and model inference should remain separable.

5 Conclusion

Productive friction must remain selective. Cognitive forcing can reduce over-reliance but may be disliked [2]. Designers should scale both friction and cross-sensory detail with uncertainty, consequence, and reversibility. Every cue needs an accessible alternative; haptics should support access and embodiment without becoming mandatory evidence.

This paper is intended to provoke further discussion and debate rather than present an evaluated design framework; it raises two empirical questions:

1. How should sensory detail scale with model uncertainty?

2. How should friction scale with an operation's scope and reversibility?

As visualization becomes more malleable and perceptually fluent, designers should dedicate effort to preserving directness (and thus, perceived user agency), expose AI limits, and prevent sensory detail from overstating model certainty. This way, interfaces of the future can make use of human sensory perception in a way that conveys AI mediation more realistically instead of exploiting multisensory perception to reinforce weak, uncertain, or inaccurate claims.

Figure Credits

Figure 2 was created by the author for this paper. Figure 1 uses film frames for scholarly criticism and analysis. Panel (a): *Iron Man 2* (2010), © 2010 MVL Film Finance LLC; source: Ortiz [12]. Panels (b–c): *Blade Runner 2049* (2017), © 2017 Alcon Entertainment, LLC; sources: Territory Studio via Failes [6] and fxguide [15]. Composition and labels are by the author.

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